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PROJECT PHASE 1:

Introduction (Proposal Stage)

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Project Title

Analyzing the Impact of Logistics Performance on Customer Satisfaction and Revenue Trends in Brazilian E-Commerce

Problem Background

The e-commerce industry requires a comprehensive understanding of the entire transaction lifecycle, encompassing customer actions, order processing, and logistics. Operating within the Brazilian market, platforms like Olist must manage complex interactions involving diverse customer locations, specific product catalogs, dynamic shipping rates, and exact geolocation coordinates for buyers and sellers.

Operational efficiency is heavily reliant on delivery performance, yet it can be difficult to measure how logistical bottlenecks such as prolonged shipping durations or late deliveries directly influence negative customer reviews and overall satisfaction. Additionally, businesses face challenges in accurately tracking seasonal purchasing behaviors and identifying which specific geographical regions are driving sales volumes versus those experiencing delivery delays.

By implementing a two-tier Business Intelligence architecture using Alteryx Designer for ETL processes and Tableau or Power BI for analytics, raw fragmented data can be transformed into a cohesive analytical model. Developing interactive dashboards will empower users to explore sales trends dynamically, evaluate the operational impact on customer satisfaction, and make data-driven decisions to optimize regional logistics.

Objectives

- To investigate and visualize the correlation between logistics performance (prolonged shipping durations) and customer satisfaction review scores.
- To analyze seasonal purchasing behaviors and track monthly and quarterly revenue trends or spikes across the platform.

- To perform geographic segmentation to identify which Brazilian states generate the highest sales volume and compare them against states experiencing the longest delivery delays.
- To identify the top-performing product categories and evaluate the specific sellers driving the highest sales volume across the platform.
- To design and execute an automated ETL (Extract, Transform, Load) workflow using Alteryx Designer to clean, standardize, and integrate five fragmented e-commerce datasets into a single cohesive data model.

Datasets Description

This project uses 5 sets of interconnected data from the Brazilian E-Commerce Public Dataset by Olist, which offers a holistic understanding of e-commerce, including customer actions, order processing, and logistics in Brazil.

1. **Dataset 1: Customers Dataset**

   This dataset identifies unique customers and their physical locations across Brazil. It is used to track repeat purchase behaviour and performing regional demographic analysis for understanding of the customer base.

   - **Source:** Kaggle - Brazilian E-Commerce Public Dataset by Olist
   - **Type of Data:** Structured (CSV)
   - **Key Variables and Description:**

Key Variables	Description
customer_id	Unique identifier that links to orders dataset.
customer_unique_id	Unique identifier for a customer across multiple orders.
customer_zip_code_prefix	First five digits of the customer's zip code.
customer_city	Name of the city where the customer is located.
customer_state	State abbreviation such as “SP” or “RJ”.

2. **Dataset 2: Orders Dataset**

As the core dataset, it records the complete lifecycle and status of every transaction made on the platform. It shows critical timestamps for purchase, approval, and delivery, which are used to measure operational efficiency. This data is the main source for analysing delivery performance and order fulfilment trends.

- **Source:** Kaggle - Brazilian E-Commerce Public Dataset by Olist
- **Type of Data:** Structured (CSV)
- **Key Variables and Description:**

Key Variables	Description
order_id	Unique identifier for each order.
customer_id	Link to the customer dataset.
order_status	Status for each order.
order_purchase_timestamp	Date and time when the purchase was made.
order_delivered_customer_date	Actual delivery date to the customer.
order_estimated_delivery_date	Estimated delivery date promised at the time of purchase.

3. **Dataset 3: Order Items Dataset**

This dataset shows a full view of every product sold, including its price and associated freight costs. It allows for detailed analysis and identifies which sellers and products drive the most volume. Besides that, it also acts as the bridge between order transactions and specific product catalogue details.

- **Source:** Kaggle - Brazilian E-Commerce Public Dataset by Olist
- **Type of Data:** Structured (CSV)
- **Key Variables and Description:**

Key Variables	Description
order_id	Link to the orders dataset.
product_id	Unique identifier for the product.
seller_id	Unique identifier for the seller.
price	Item price for each unit.
freight_value	Shipping cost for the item.
shipping_limit_date	Seller's deadline for handling the order.

4. **Dataset 4: Order Reviews Dataset**

   This dataset contains customer feedback and satisfaction scores, which includes both numerical ratings and raw text comments that can be used for service analysis. Linking this to order data helps identify if logistics issues like late deliveries impact customer satisfaction.

- **Source:** Kaggle - Brazilian E-Commerce Public Dataset by Olist
- **Type of Data:** Structured and Unstructured (CSV)
- **Key Variables and Description:**

Key Variables	Description
review_id	Unique identifier for the review.
order_id	Link to the orders dataset.
review_score	Rating ranging from 1 to 5.
review_comment_message	Text content of the customer's review. It is an unstructured component within a structured file
review_creation_date	Date the survey was sent.

5. **Dataset 5: Geolocation Dataset**

   The dataset gives exact latitude and longitude data for Brazilian zip codes that are utilized by buyers and sellers. It is useful in determining shipping distances, as well as developing geospatial maps including sales heat maps.

- **Source:** Kaggle - Brazilian E-Commerce Public Dataset by Olist
- **Type of Data:** Structured (CSV)
- **Key Variables and Description:**

Key Variables	Description
geolocation_zip_code_prefix	Link to customer/seller zip codes.
geolocation_lat	Latitude coordinates.
geolocation_lng	Longitude coordinates.
geolocation_city	Name of the city.
geolocation_state	State abbreviation.

Proposed BI Approach

1. Tool Utilization Strategy

The proposed Business Intelligence solution will utilize a two-tier architecture by leveraging Alteryx Designer for backend data preparation and a visualization tool such as Tableau or PowerBI for front-end analytics.

Alteryx Designer will serve as the core data engine that is responsible for all Extract,Transform, and Load (ETL) processes. It will be used to ingest the raw CSV files specifically the orders, items, reviews, and customer datasets. We will use Alteryx Designer to standardize the formats and blend all of it into a single cohesive relational dataset. For the last step, the necessary aggregations are performed to ready the data for visual analysis.

Once the data is thoroughly cleaned and prepped, it will be imported into a visualization tool, such as Tableau or Microsoft Power BI. Within this front-end platform, we will build interactive dashboards featuring dynamic slicers such as date ranges and geographic regions that empower stakeholders to intuitively explore sales trends as well as pinpoint logistics bottlenecks and analyze customer satisfaction metrics.

Tools Used	**Purpose**	**Description**
Alteryx Designer	To create data workflow and ETL (Extract, Transform, Load)	Act as the data engine. It will ingest the raw CSV files and standardize the formats which will be combined into a single cohesive dataset. Aggregations can be performed upon the dataset for visual analysis.
Tableau / Power BI	To create dashboard with interactions	The cleaned dataset from Alteryx Designer will be imported to be used for creation of interactive dashboards with slicers to allow users to explore sales trends and customer satisfaction.

2. Data Workflow

To ensure accurate and meaningful analysis, the datasets will undergo a rigorous preparation process within Alteryx Designer.

Phase	**Tools**	**Methodology**

Data Integration	Join Tool	• Merge the Orders, Items, and Reviews datasets using the common order_id key. • Join the Customers dataset using customer_id to link geographic data to each transaction.
Data Cleaning	Data Cleansing Filter Tool DateTime	• Filter out incomplete records and remove duplicate rows. • Convert string-based timestamps into actual date formats for accurate time-series analysis.
Data Transformation	Formula Tool Summarize Tool	• Calculate Shipping Duration • Group data by customer state and product category to calculate total revenue, average shipping times, and satisfaction scores.

3. Analysis and Visualisation Strategy

The project will focus on identifying correlations between logistics performance and customer satisfaction, as well as tracking revenue trends across different regions.

To thoroughly understand the business problem, our team will conduct three primary forms of investigation. First, we will perform a trend analysis to track seasonal purchasing behaviors and identify any monthly revenue spikes over the dataset's timeframe.

Next, we will execute geographic segmentation to determine which Brazilian states drive the highest sales volumes and which specific regions struggle with the longest delivery delays. Finally, we will run a correlation analysis to explicitly investigate the relationship between prolonged shipping durations and negative customer review scores.

Appropriate Visual Forms effectively communicate these insights to stakeholders. We will leverage five specific visual formats in our final dashboard. We will use a line chart to clearly visualize revenue trends and order volumes over monthly and quarterly timeframes.

A geospatial map will be implemented to display the concentration of customers and total sales distributions across various Brazilian states. To illustrate our correlation findings, a scatter plot will map Shipping Duration on the x-axis against Customer Review Score on the y-axis. We will also rely on a bar chart to compare the top 10 best-selling product categories or rank the states experiencing the highest average delivery times.

Finally, we will feature KPI text cards to instantly highlight high-level metrics at a glance, specifically focusing on Total Revenue，Average Delivery Time and Average Satisfaction Score.